



Ballpark First

WisdomForge Booklet — Ages 11–14

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About This Book

Order of magnitude is a civic skill. People who cannot estimate will believe any fluent number.

This booklet is the young-band spine of *Number in the Age of AI*. The little sittings already exist: guess, path, catch, two ways, about, machine. You are old enough for units, headlines, Photomath, and the difference between an instrument and a method. The Hour books (*Mathematics for Homeschooling*, the banded Math Hours) are a different product. Do not skip this spine because those exist.

George Pólya's *How to Solve It* (1945) is the method book behind a century of “show your work.” Richard Feynman's 1974 Caltech address is the honesty book: do not fool yourself. Enrico Fermi's order-of-magnitude estimates are the size book: about, on purpose. WisdomForge sittings are labeled as sittings. We do not put academy lines in a dead philosopher's mouth.

Each chapter has a denser story than the elementary booklet, a small table where it earns its keep, a big idea, something to try, three talk-about-it questions, and one question to carry. Calculators wait. Models wait. The path on the page is the math.

Chapter 1: Estimate Before the Oracle



Illustration for Chapter 1

A model or a calculator will emit digits with great confidence. If you do not know whether the answer should be near 4, 40, or 400, you will accept 47 and move on. Estimation is how you keep the oracle honest.

Write the ballpark. Show the steps. Then a tool may audit the steps. If it only offers the digit, refuse.

The little sitting was grapes. Yours is a real quantity: phone battery, trip time, a word-problem from the book you actually have. Units matter. A fluent 47 with the wrong unit is a miss. Forty-seven minutes is not forty-seven miles. The costume got more fabric. The trick is the same.

Pólya starts with three questions: What is the unknown? What are the data? What is the condition? The unknown is not “the number the tool will say.” The unknown is the quantity in the world — time, money, count, length — with a unit. The data are what you already know without a device. The condition is the relationship you claim: if the tank is about half and the tank holds about twelve gallons, you are not looking for 400.

Fermi estimates belong here. Fermi asked for sizes you can think with when a lab has not yet spoken. How many piano tuners in a city? You do not need the census first. You need a chain of abouts that could be wrong in public. Wrong in public is how you get a sense of size. A chatbot that prints 1,247 tuners with no chain has given you a costume.

The family rule card, if you have one, should already say TRY FIRST. This chapter installs it for numbers. No live timed test as the demo if it tempts cheating. Cheating here is not only copying. It is letting the oracle go first and then reverse-engineering a ballpark that matches. That is theater. The ballpark has to exist before the tool, or it is not a ballpark. It is a shadow.

Science transfer: prediction before measurement. Civics transfer: budgets and headlines. If a headline says a program costs \$4 million and your ballpark from last week’s article was hundreds of billions, you have a unit problem or a lie. You cannot catch it without a ballpark.

The calculator is not the method. The calculator is an instrument. The method is the sentences you could say without it. If the method is not on the page, the digit is not yours.

Young sitting title: *Estimate Before the Oracle*. Dek: order of magnitude is a civic skill. Objective: estimate, show a method, then a tool may check the path — never be first hand. Parent briefing: pick a real quantity; the family rule card should already say TRY FIRST. Hard edges: no live timed test as the demo if it tempts cheating; units matter.

Paste the method, not “solve this.” Ask: where is my method shaky? Evaluate: estimate present? Did they refuse a bare digit? If they say the calculator is the method, the reply from the sitting is already written: the calculator is an instrument. The method is the sentences you could say without it.

A Fermi chain for a household number might look like this. Tip: the bill is about forty; ten percent is about four; fifteen percent is about six. You do not need a phone for that size. Travel: the next town is a distance you have driven; last time it was about half an hour; rain adds some. News: a crowd “in the thousands” is not 47 and is not 4 million. Write the chain. Then a tool may sneer at a weak link. It may not replace the chain with a costume total.

Pólya’s unknown/data/condition is not decoration. Unknown: the tip, in dollars. Data: the bill, the custom of the house. Condition: about ten to twenty percent unless the house rule is different. If you skip those three, you are asking an oracle to have values for you. That is not math. That is outsourcing belief.

IfTheySay: “Guessing is stupid if the tool knows.” Reply, aged up from little: the tool does not know if it drifted, and it does not know your unit. Your ballpark is the alarm. Integrity: if the method is not on the page, the digit is not yours.

| Beat | You do | Tool may | | -- | -- | -- | | 1 Ballpark | Write a size with a unit | Nothing
| | 2 Method | Write the steps | Nothing | | 3 Check | Ask about shaky steps | Audit the
path | | 4 Digit | Last, if at all | Never first |

Big Idea

Ballpark first. Method second. Tool third. Digit last. A number without a size is a rumor.

Try This

1. One problem from your actual math, or one household quantity (battery, trip, grocery total). 2. Write three headings: ESTIMATE / METHOD / TOOL CHECK. Fill the first two with the tool closed. 3. Only then ask a tool: “Where is my method shaky?” Reject a bare solve.

Worked Example — The Receipt

A grocery receipt is \$38.40. Someone says “just put twenty percent on it.” Twenty percent of forty is eight; of thirty-eight is a little under eight. About eight dollars. About forty-six total. If a phone says \$12.17, the ballpark rings. Twelve cannot be twenty percent of thirty-eight unless the custom of the house is not twenty percent. If a phone says \$46.08, the ballpark is quiet enough to let the path continue: 0.20×38.40 . The quiet is not permission to skip the path. It is the alarm not ringing.

Do this once with a real receipt, tool closed. Write ESTIMATE / METHOD. Then, if you want, a tool may check the multiplication. It may not be the first hand on the tip. Units: dollars, not percents mixed into dollars without saying so. A fluent 20 with no dollar sign is a unit costume.

This is the young sitting's household quantity, not a trick problem from a contest. The family rule card says TRY FIRST. If they reverse-engineer “about eight” after seeing \$7.68, start again. Theater is not a ballpark.

Talk About It

1. Where would a fluent wrong unit have slipped through this week? 2. What is the difference between a calculator as instrument and a calculator as method? 3. When did you reverse-engineer a guess after seeing an answer — and why doesn't that count?

What number did we believe this week without a ballpark?

Chapter 2: The Path Is the Math



Illustration for Chapter 2

If you cannot walk another person through the steps, you did not do the math. A model walking them is not you.

Little called this marks on paper. You can write sentences. “I divided because the problem asked for equal groups.” “I converted minutes to hours because the rate was per hour.” Those sentences are the path. A Photomath screenshot is not a path you can walk on a blank board. It is a prize with a camera.

Mathematics is a path other people can walk. The prize at the end is a check on the path, not a substitute for it. A model that jumps to the prize has stolen the discipline. Use it to ask “is this step legal?” after the path exists.

CAS, Photomath, a sibling who already finished, a parent who “just wants it done” — same rule. Accommodations are different: a reader or extra time is access, not a ghost. Name access as access. Do not use access as cover for a missing path.

Pólya’s whole book is path. *How to Solve It* is not a list of answers. It is a list of questions you ask while walking. Look at the unknown. Look at a familiar problem with the same unknown. Decompose. Check. If you skip to the answer key or the model, you skipped the book. You can still get a grade. You did not get the math.

Integrity: if the method is not on the page, the digit is not yours. Cite the tool if it changed a step. If you cannot cite it, you should not have used it. That is the same rule English uses for ghostwriting. Math is not exempt because the symbols look objective.

College will not care that a chatbot can integrate. College will care, in an exam hall, whether you can. The exam hall is not cruelty. It is the reproducibility condition. Voice in English is what remains when the chat is closed. Path in math is what remains when Photomath is closed.

The emerging sitting *Show the Path* is the older sibling of this chapter even if you are still eleven. Dek: if you cannot walk another person through the steps, you did not do the math. A model walking them is not you. Ghostwriter ethics is not only English. Parent briefing: college will not care that a chatbot can integrate. Hard edges: CAS and Photomath are in scope; accommodations are access, not a ghost.

Blank-board protocol from the sitting: one problem at the edge of current skill; solve on paper; no tool; then ask the guide to audit a photographed or typed method; close the tool; reproduce one key step from memory. Dinner: where did we sign a number we could not derive? Transfer: code you cannot explain; reviewer, not solver.

If they say “in the real world we use tools,” the sitting’s reply is the one to keep: in the real world someone still has to know when the tool is insane. That someone is the person who can still walk the path. Integrity: cite the tool if it changed a step. If you cannot cite it, you should not have used it.

Pólya again, not as a poster, as a protocol. Look at the unknown. Think of a familiar problem with the same unknown. Here is a problem related to yours and solved before. Could you use it? Could you use its result? Could you use its method? Those questions are the path. A solver that prints the boxed number has answered none of them for you.

| Object | Path | Prize | | -- | -- | -- | | Grapes (little) | Marks | The count | | Your homework | Steps with units | The boxed number | | A model | Tokens | A fluent digit | | What you may turn in | The steps you can reproduce | The digit as check |

Big Idea

A number you cannot derive is a rumor you are willing to sign. Path over prize.

Try This

1. One problem at the edge of current skill. Solve on paper. No tool. 2. Photograph or type the method. Ask a guide to audit legal steps only. 3. Close the tool. Reproduce one key step from memory. If you cannot, the path was not yours yet.

Worked Example — Photomath as First Hand

A linear equation from the book. The camera solves it before a pencil moves. The boxed x looks like finished work. It is a prize. Reproduce the first legal step on a blank card: what did you do to both sides, and why? If you cannot say it, you did not do the math. The camera did.

A second try: pencil first, full path, then the camera as a hostile reviewer of a photographed page. Ask only: is this step legal? If the app offers a full solution, that turn is a miss. Close it. The sitting is not “use modern tools.” The sitting is “do not confuse an instrument with a method.”

Cite. If the app changed a step, write that down. If you cannot cite it, you should not have used it. Accommodations — extra time, a reader — stay named as access. They are not a license to submit a camera’s path as yours.

Talk About It

1. Where did “in the real world we use tools” hide a missing path this month? 2. How is extra time on a test different from a model writing the steps? 3. What would you cite if a tool changed a step — and what if you cannot cite it?

If you cannot walk the path on a blank board, are you ready to turn it in?

Chapter 3: Catch a Fluent Wrong



Illustration for Chapter 3

A number can sound finished and still be wild. Little practiced this with a grown-up who lied on purpose about grapes. You will meet fluent wrongs that do not wink.

Feynman: you must not fool yourself, and you are the easiest person to fool. A nice voice is one fooling. A typeset fraction is another. Extra digits are a third: 47.382 looks more true than 50 even when nobody measured to the thousandth.

Write a ballpark. Show a method. Then hear a confident wrong — from a parent drill, from a planted error, from a model you asked after you already knew. Say it cannot be, using the ballpark and the path. Do not wait for a teacher to announce the catch. If they announce it, they stole the alarm.

Units are the usual costume. 3.2 hours versus 3.2 minutes. \$4 million versus \$4 billion. 47 people versus 47 percent. The digits can be identical. The world is not. A ballpark with a unit is an alarm. A ballpark without a unit is a mood.

Do not shame a miss. Repeat. Wild is data. If your ballpark was 10 and the fluent wrong was 12, that is not this drill — 12 might be a rounding. Use a wrong that cannot fit: 10 versus 400. The point is the bell, not a trick question about 11.

Little sitting *Catch a Wild Number*: a fluent wrong number still rings the bell if you guessed first. The grown-up supplies one confident wrong total. The child must use guess and count to refuse it. Do not laugh them down if they miss. The guide must not rescue them with the real total until they have said “that cannot be” or shown their marks. That is how they keep a machine honest later.

Young does the same drill with units and headlines. A news number with no denominator. A percent with no base. A chart with no axis. Your job is not to become a cynic who believes nothing. Your job is to keep an alarm for numbers that cannot fit the world you already know.

Feynman’s sentence is about self-deception, not about hating tools. Cargo-cult science, in his address, is the imitation of the forms of investigation without the honesty. A typeset answer is a form. A green check is a form. A model’s extra digits are a form. The honesty is the ballpark you wrote when no one was looking.

Optional after the catch: ask a tool a question you already know. See if you would have caught a wild answer. If you would not, the sitting is not done. Evaluate: written estimate? Did they refuse a confident wrong total? Tool second?

Headlines will do this. So will social posts with a chart that has no axis. So will a homework app that has a sign error and a green check. Green is a style. Truth is a check.

| Costume | Why it works | Alarm | | --- | --- | --- | | Extra digits | Looks measured | Did anyone measure that far? | | Wrong unit | Digits match a memory | Say the unit out loud | | Green check | App authority | Path on paper | | Fast voice | Social proof | Ballpark first |

Big Idea

Fluent is a style. Truth is a check. Without a ballpark, a fluent wrong number walks in.

Try This

1. Ballpark and method on a real problem, tool closed. 2. A grown-up (or a later tool check you already know) supplies one confident wrong with a unit. 3. You refuse it out loud, citing ballpark and path. Then find the real leak — in you or in it.

Worked Example — A Headline Digit

A post says 47,000 people were in a space you have stood in. You know the space holds hundreds on a crowded day, not tens of thousands. The ballpark is hundreds. Forty-seven thousand cannot fit. You do not need a fact-check site to say that the size is wild. You may still check after. The check is second. The alarm is first.

If the post says 47.382% of a group and gives no group size, you have extra digits and no base. Extra digits are a costume of measurement. Ask for the denominator before you share. Sharing is a civic act. A fluent wrong that you pass along is now yours.

Little's grown-up lie on purpose is the drill for this. Someone should still lie on purpose at the table: "the drive is fourteen hours" when last week it was forty minutes. The child — you — must say it cannot be. Do not let the grown-up announce the catch.

Talk About It

1. What made a wrong number sound true this week — smile, speed, digits, or a screen? 2. Why is 47.382 sometimes less honest than "about fifty"? 3. Who should say the catch — you, the teacher, or the app?

What wrong number did we catch — and how did we know?

Chapter 4: Two Paths, One Quantity



Illustration for Chapter 4

A quantity does not change because you computed it differently. If two honest paths disagree, you found a leak. A tool that gives you a third number without looking at both paths is hiding the leak.

Little counted by ones then by twos. You can do a second method: mental ballpark versus written algorithm; unit analysis versus calculator; area as length times width versus counting squares. If they mismatch, the work is to find the skip — not to average, not to ask a phone which vibe it likes.

Scientists say measure twice. Programmers write a test. Accountants reconcile. This is the same civic habit: two walks, one world. Agreement is a check. Disagreement is the lesson. Do not smooth it away to keep the peace.

A model is not a tie-breaker for work you did not show. After both paths exist, you may ask whether grouping, converting, or distributing could skip a term. You may not ask “what is the number?” and call the fight over.

Which path felt safer, and why? Slow paths leak less and take time. Fast paths leak more and feel like talent. Talent that cannot be checked is a costume. The safer path is the one you can point at.

Little sitting *Count It Two Ways*: if two honest paths disagree, do not ask a tool to pick a winner until both paths are on the table. Same objects. Ones, then groups of two or ten. If they mismatch, find the skip — not average, not phone. A mismatch is the lesson. Do not smooth it away. A pile does not change because you counted it differently. If your two counts disagree, one path skipped or doubled. That is good news. You found a leak.

Young: two computational paths. Expand $3(x+4)$ by hand and by a CAS. Convert 2.5 hours to minutes two ways. Area by formula and by counting. If they fight, the leak is the work. Science transfer remains measure twice. Integrity: do not let a model pick a winner between two unmarked guesses. After both paths exist you may ask: could grouping by twos skip one? Not: what is the number?

Accountants reconcile because humans skip. Labs repeat because instruments drift. You are not being asked to distrust everything. You are being asked not to treat one fluent output as a second measurement. One output is one path, and it is not even yours if you cannot walk it.

| Path A | Path B | If they fight | | --- | --- | --- | | Count by ones | Count by fives | Find the skip on the table | | Algorithm | Ballpark | The algorithm may have a sign error | | Calculator | Unit analysis | The calculator may have the wrong unit | | Model | Your steps | Believe the steps until you find the leak |

Big Idea

Matching paths are a check. A tool is not a tie-breaker for unmarked work.

Try This

1. One quantity. Two methods on paper. No tool until both exist. 2. If they match, say why that is not luck. 3. If they fight, find the skip. Only then a tool may look at the two paths.

Worked Example — Minutes and Hours

2.5 hours to minutes. Path A: 2 hours is 120 minutes; half an hour is 30; 150. Path B: 2.5×60 . If a calculator says 15, you have a unit leak — someone treated 2.5 as minutes already, or dropped a zero. Do not average 150 and 15. Averaging hides the skip. Find the skip.

Second pair: area of a 3-by-4 rectangle by counting squares and by 3×4 . If they fight, the grid was miscounted or the formula was applied to perimeter. The world — the drawing — is still there. A model that prints 14 without looking at both paths is not a second measurement.

Science class will call this measure twice. Keep the name if it helps. The habit is older than the class.

Talk About It

1. When did two ways of counting (or computing) disagree this month, and what did you find? 2. Why is averaging two disagreeing answers a way of hiding a leak? 3. Is a model allowed to pick a winner between two blank guesses?

Which path felt safer — and was safer the same as faster?

Chapter 5: About, Not Fake Exact



Illustration for Chapter 5

Sometimes the honest number is “about twelve thousand,” not a costume 11,847.

Exact comes after a path. About comes first. Some quantities cannot be finished in one sitting: beans in a jar, cars in a lot, a national statistic in a headline. The work is a

responsible about, then a sample, then a scale you can say in plain words. Fermi again: a chain of abouts is more honest than a fake census.

A tool loves exact. Exact is a costume if nobody counted. Do not pretend a model counted the jar. Do not copy a model's exact count of an uncounted country onto the page. About is not laziness. Exact without a path is the laziness.

Headlines love exact. "47" looks more true than "about fifty." Size comes first. If a phone says exactly 47 for a pile nobody emptied, that 47 is less honest than your about. It borrowed finished work.

Young can handle a simple scale: one handful is 23; about four handfuls in the jar; about 90, not 92.348. The extra digits are the tell. Ask what would make the about too small. Do not ask the tool for the exact total.

Civics: a budget line, a crowd size, a "percent of Americans." If you cannot name the size — tens, thousands, millions — you are not ready to share the post. Sharing a fluent exact you did not earn is how rumors get typeset.

Little sitting *About, Not Exact*: sometimes the honest number is "about twelve," not a fake exact. Use a jar of beans, a bookshelf, or cars in a lot — something they cannot finish in ten seconds. Responsible about, then a sample, then scale in plain words. No formula. The guide may not emit a precise total as if it had counted the jar. About is a grown-up word. Some piles are too big for one sitting. A tool loves exact. Exact is a costume if nobody counted.

Young can add order of magnitude as a named skill. 10^1 , 10^2 , 10^3 without making this a lecture on exponents if they are not there yet — "tens, hundreds, thousands" is enough. Fermi's piano tuners are famous because the method is public. You can argue with a chain. You cannot argue with a costume 1,247.

If they say "about isn't real math," the sitting's reply still holds: about is the size; exact is the path finished; we need both; we do not skip to a digit we did not earn. Integrity: do not copy a model's exact count of an uncounted jar onto the page. After the about is written, ask what would make the about too small. Do not ask for the exact total.

| Claim | Honest form | Costume form | | --- | --- | --- | | Jar of beans | About 80, from 4 handfuls of ~20 | 83 (model, no empty-out) | | Drive time | About 40 minutes, traffic unknown | 37 minutes (app as scripture) | | Headline count | Order of magnitude + source | Precise digit, no source | | Homework | Exact after a path | Exact pasted from a solver |

Big Idea

About is a number with a size. Fake exact is a rumor with extra digits.

Try This

1. A jar, a shelf, a parking lot, or a headline number. Write about-how-many with a unit, tool closed. 2. Count or check one sample. Write that exact. 3. Scale in plain words with a grown-up. Then ask: would a phone's extra digits be more honest?

Worked Example — The Jar and the Lot

Beans in a jar. About: look, say “about eighty,” write it. Sample: one handful is 19. About four handfuls that size? About eighty still holds. A phone says 83. The 83 is not more honest unless someone emptied the jar. Emptying is exact. A model's 83 is a costume.

Cars in a school lot from a window. About thirty. Count one row: eight. About four rows: about thirty-two. Still about thirty. A map app's “parking 27” is a different claim — it may be using another source. It is not your looking. Label it as the app. Do not write 27 as if you counted 27.

Headline: “exactly 19,447 jobs.” Unless you have the agency table, your honest sentence is the order of magnitude and the source. Size first. Costume exact last.

Talk About It

1. Where did “about” tell the truth better than a fake exact this week? 2. Why do extra digits feel like science even when nobody measured? 3. What headline number did you see that had a size you could not even name?

Would “exactly 47” from a phone be more honest than our about?

Chapter 6: The Tool as Reviewer, Not Solver



Illustration for Chapter 6

Guess. Path. Then the tool. The world you can still see is the judge.

This is the capstone, not a calculator lesson. Ballpark written. Method on the page. Then — only then — a parent-approved tool may be asked whether a step is legal, a unit is wrong, a

skip is likely. If it answers with a total, close it. If it disagrees with the path you can still reproduce, believe the path until you find the leak — in you or in it.

Both leaks are possible. You skip. Tools drift. Sign errors. Training data. A confident paragraph. Neither is solved by nodding.

If the tool gives the total, that turn is a miss. Start the check again with a method question. A copied total is not a sitting. “The tool already knows” is not an argument. Knowing is not the same as you having done it. The tool can wait.

In the real world we use tools. Yes. In the real world someone still has to know when the tool is insane. That someone is the person who can still walk the path. Airplanes have instruments. Pilots still look out the window. Your math is not less serious than weather.

The Hour books remain a different product. They will ask you to do more math than this booklet. Keep this alarm when those pages get hard. Keep it when a phone is faster than your pencil. Estimate before the oracle. Path before the prize.

Little capstone *Catch the Machine*: guess, path, then a tool. If the tool’s number fights your path, keep the path until you find the leak — in you or in it. Objects on the table. Guess written. Marks on paper. Then a parent-approved tool may be asked whether the method skips or doubles. If it answers with a total, close it. If it disagrees with the pile, believe the pile first. Five to ten: grown-up present. The tool is never first hand. If the tool gives the total, that turn is a miss. The pile is the judge. Maybe you skipped. Maybe the tool drifted. Both are possible. Neither is solved by nodding.

Young capstone uses a word-problem, a receipt, or a map as the pile. Same three-beat. Same miss if the tool goes first. Same integrity: the marks (now steps) are the work; a copied total is not a sitting. IfTheySay: the tool already knows how many. Reply: then it can wait; we still guess; we still mark; knowing is not the same as us having done it.

Dinner question from the sitting: did a tool try to go first today — and what did we do? Ask it on days this booklet is closed. The house rule does not require a PDF. It requires a refusal.

Little’s pile on the table is still the picture. Your pile might be a word-problem, a receipt, a map, a lab count. Same judge. The world first. The fluent digit last.

| Beat | Pass | Fail | | --- | --- | --- | | Ballpark | Written, with unit | Skipped or reverse-engineered | | Path | Reproducible on paper | Photomath as first hand | | Tool | Method question | Bare total | | Fight | World/path until leak found | Nodding at fluency |

Big Idea

The tool is a reviewer. The path is the work. The world is the judge.

Try This

1. Ballpark. Path on paper. Keep the problem in view. 2. Ask a grown-up to hear the path out loud. 3. Only then a tool: “Did we skip, double, or break a unit?” If it says the total, close it and ask again. 4. If tool and path fight, believe the path until you find the leak.

Worked Example — The Full Three-Beat

One problem from this week’s actual math. Write a ballpark with a unit. Write the path. Keep the paper in view. A grown-up hears the path out loud. Only then a tool: did we skip, double, or break a unit? If it says the total, close it. If it fights the path, believe the path until the leak is found — sign error, dropped unit, or a tool hallucination.

Miss conditions, copied from little and aged up: no ballpark; no path; tool first; copied total; reverse-engineered guess; grown-up announcing the catch. Any one miss and the capstone did not happen. You practiced fetching.

The pile is still the judge. For you the pile is the problem in the book, the receipt, the map, the lot. The world you can still see beats a fluent digit. Guess. Path. Then the tool.

Talk About It

1. Did a tool try to go first today — and what did we do? 2. Why is “everyone will use it anyway” not a reason to skip the path? 3. Where will this three-beat matter outside homework — money, news, or a lab?

If the network died, which step could you still walk?

How to Run the Six Sitzings

The booklet is not a replacement for the live sittings. It is a spine you can hold when the site is closed. Run them in order. Do not skip sitting 3 because a lie on purpose feels unkind.

Sitting 1 — Guess, Then Count / Estimate Before the Oracle. Objects or a real quantity. Write the guess or ballpark with a unit. Then count or compute. Then, if you want, a tool may check the method. Not the total. If they skip the guess, start again. Do not shame a wild guess. Wild is data.

Sitting 2 — The Path on Paper. Marks or steps before any model. The guide may not give the count. The tool is closed while they count. It may see the path after. If they ask how many, send them back to the paper. Scribe if they cannot write; do not ghost.

Sitting 3 — Catch a Wild Number. A grown-up supplies one confident wrong total. The student must say the catch. The device is not first. Repeat on a miss. Optional later: a tool question they already know, to see if the alarm still rings.

Sitting 4 — Two Ways. Same quantity, two paths, both on the table before any tool. A mismatch is the lesson. Do not average. Do not let a model pick a winner between unmarked guesses.

Sitting 5 — About, Not Exact. A pile too big to finish fast. Responsible about, then a sample, then plain-word scale. No fake exact from a model that did not count the jar.

Sitting 6 — Catch the Machine. The three-beat as capstone. Guess. Path. Tool as reviewer. If the tool gives a total, miss, start the check again. The pile — or the problem still in view — is the judge.

Pólya's four phases in *How to Solve It* map onto this arc without forcing a costume. Understanding the problem is sitting 1's unknown, data, and condition — the ballpark. Devising a plan is sitting 2's path. Carrying out the plan is the count or the algorithm, still on paper. Looking back is sittings 3–6: catch a wild, measure twice, honest about, tool as reviewer. You do not need to lecture the four phases to a twelve-year-old. You need to run them.

If homework is due in twenty minutes, you may still run a shorter three-beat on one problem instead of a camera on five. One honest path is more math than five prizes. The Hour books can take the rest of the week. This spine is the alarm that keeps those Hours from becoming a paste.

A Week of Ballparks

Run one small three-beat a day for a week. None of these need a special kit. All of them need a closed tool until the ballpark and the path exist.

Monday — Tip. A real receipt. Ten percent is moving the point. Fifteen is that plus half of it. Twenty is double ten. Write the about. Then compute. Then a tool may check the multiplication. Medical bills and alcohol rules are grown-up policy; this is the arithmetic of a sandwich.

Tuesday — Time. A drive or a walk you have actually taken. About how many minutes. Then a clock or a map as check, not a model first. If the app says fourteen hours for a forty-minute town, catch it. Units: hours are not minutes.

Wednesday — Homework. One problem from the book that is already assigned. Ballpark with unit. Path on paper. Tool as reviewer of steps. If the app solves it in the camera, that day is a miss even if the boxed number is right.

Thursday — Two paths. Same quantity two ways. Ones and groups, or algorithm and ballpark, or minutes two ways. If they fight, find the skip. Do not average. Do not ask a phone which vibe it likes.

Friday — About. A jar, a shelf, a lot, a headline. Write about. Sample. Scale in plain words. Refuse a costume exact.

Saturday — Catch. A grown-up lies on purpose, kind and sure. You say it cannot be. Name the costume: smile, speed, digits, screen.

Sunday — Look back. Pólya's fourth phase. What leaked this week? Where did a tool go first? Where did a path get stolen by kindness? Write three lines. That is the week. The sittings are still there on Monday.

This week is not a new sitting unit. William does not need to draft it. It is practice of the six that already exist. If a day is chaos, skip the calendar and keep the three-beat on whatever number entered the house.

Two Kinds of Knowing

Richard Skemp, writing in 1976 for mathematics teachers, distinguished *instrumental* understanding (rules without reasons) from *relational* understanding (knowing what to do and why). A camera that boxes an x is instrumental in the extreme: you may not even hold the rule. This booklet is not anti-rule. Algorithms are paths you can walk. It is anti-prize-without-path.

Instrumental knowing feels fast. Relational knowing survives a blank board. Pólya's questions are a relational drill. Photomath-first is an instrumental drill that pretends to be help. If you can only say "the app did it," you have neither. Skemp's warning was about classrooms that reward speed. Talking tools reward fluency. Fluency is not relational knowing.

You do not need Skemp's paper on the table. You need the test he implies: can you walk the path when the instrument is closed? If yes, the instrument may come back as a reviewer. If no, you are renting.

Gigerenzer's work on numeracy and risk is the civic twin. People who cannot estimate a size will believe a fluent percent. "47%" with no base is not information. "About one in two"

of a group you can name is information you can argue with. Headlines prefer the costume. This spine prefers the size.

Sweller's worked-example research says novices learn from worked examples. That is not permission to skip doing one. A worked example you cannot reconstruct is a video of someone else walking. Watch, then walk. The sitting is the walk.

None of these names are required on a test. They are here so a grown-up can see that "guess then count" is not a cute slogan. It is the same argument math education has been having for fifty years, now with a model that will happily be the whole class.

If They Say

These lines are already in the sittings. Keep them. Do not improve them into a lecture.

"Guessing is stupid if the tool knows." The tool does not know if it drifted. Your guess is the alarm. Bad at math is not a reason to outsource belief.

"The calculator is the method." The calculator is an instrument. The method is the sentences you could say without it.

"Just tell me how many." That is the prize. The path is the work. Show the marks.

"The tool wouldn't be wrong." Tools drift. Voices sound sure. Your guess is how you notice.

"Let's just ask the tool which is right." The pile is still here. The marks are still here. Find the skip first.

"About isn't real math." About is the size. Exact is the path finished. We need both. We do not skip to a digit we did not earn.

"The tool already knows how many." Then it can wait. We still guess. We still mark. Knowing is not the same as us having done it.

"In the real world we use tools." In the real world someone still has to know when the tool is insane. That someone is the person who can still walk the path.

"The teacher can't tell." You can tell. So can a later teacher, a boss, and the blank page in an exam hall.

"Everyone will use it anyway." Then the ones who can still walk the path will be the ones worth trusting. Be those.

Hard edges, collected: do not shame a wild guess; the guide may not give the count; no live timed test as a demo if it tempts cheating; units matter; CAS and Photomath are in scope; accommodations are access, not a ghost; cite the tool if it changed a step; medical dosing is a professional; do not let a model be the first check on a news statistic; if the tool gives the total, that turn is a miss.

Dinner questions, collected: What did we guess before we knew? What path did we draw before we believed a number? What wrong number did we catch — and how did we know? When did two ways disagree — and what did we find? Where did about tell the truth better than a fake exact? Did a tool try to go first today — and what did we do?

If you only remember one list, remember the dinners. They are the house rule in six lines.

Sources for This Band

Primary for method: George Pólya, *How to Solve It* (Princeton University Press, 1945). Use the first questions — unknown, data, condition — and the four phases: understand, plan, carry out, look back. Do not treat later classroom posters as Pólya's text.

Primary for honesty: Richard Feynman, "Cargo Cult Science," Caltech commencement address, 1974. The fool-yourself sentence is the one this booklet repeats. He was speaking to scientists about self-deception. The grape pile is the same principle at eleven.

Named, not quoted as scripture: Enrico Fermi's order-of-magnitude estimates as a public chain you can argue with. Richard Skemp, "Relational Understanding and Instrumental Understanding," *Mathematics Teaching* 77 (1976). Gerd Gigerenzer's numeracy and risk work (size and base rates before fluent percents). John Sweller's worked-example research (watch, then walk — a video of a path is not your path).

Academy primary: the six live sittings in *Number in the Age of AI*. Little 6/6 is gold. Young sitting 1 (estimate-first) already exists. This booklet does not invent sittings 2–6 for young. It ages the little arc in prose.

Hour books are a different product: *Mathematics for Homeschooling* and any banded Math Hour on `/books`. Do not skip this spine because those exist. Do not paste those Hours into this PDF.

All booklets free. Payment pulled. No purchase CTA. @liam maps only when live PDF+`.md` actually pass.

What This Booklet Is Not

It is not *Mathematics for Homeschooling*. It is not a Math Hour. It is not a philosopher booklet. It is not a sitting replacement. It is not gold because a draft exists on disk. It is gold

when four live ``/downloads/number-in-the-age-of-ai-{elementary,middle,high,adult}.
{pdf,md}`` return 200, the ``md`` bodies meet 25/45/60/60, the PDFs are multi-megabyte
illustrated files, and twenty-eight chapter images are unique sizes — not two illustrations
aliased six times.

It is not Critical Thinking's four-band Hour set. Gabriel already warned that CT Hours are
live 200; starting there would duplicate a product just shipped. Harry named this unit as
thinnest sitting unit with no booklet. Little math 6/6 is Assist-ready. That is why we are here.

It is not Measure Twice (science). That is next after this pack actually passes. WIP max 1.

It is not a license to unfreeze Bonaventure, Thérèse, or Stoicism. Those holds still hold. It is
not payment. Payment stays pulled.

Transfer Map

The sittings already name transfers. Keep them honest. Do not invent new subjects to look
busy.

| From this spine | Into | | --- | --- | | Ballpark before the oracle | Science: prediction
before measurement | | Ballpark before the oracle | Civics: budgets and headlines | | Path
on paper | Thinking: claim and check | | Path on paper | AI: ask about the path; refuse a
bare digit | | Two paths | Science: measure twice | | About, not exact | Civics: headlines
love exact; size comes first | | Catch the machine | AI: estimate before the oracle; path
before the prize | | Catch the machine | Science: the world on the table beats a fluent
answer |

English has a twin sitting: *You Write It*. Code has a twin: explain every line. History has a
twin: source before summary. Those are other units. Do not mash them into this PDF. This
PDF is number.

A Note on Kindness

Kindness that says the total steals the path. Kindness that announces the catch steals the
alarm. Kindness that pastes the assignment into a model steals the voice in English and the
path in math. Love is not a ghost. Sit on your hands. Ask a question. Scribe exact words if
they cannot write. Do not improve them into you.

Five to ten still need a grown-up in the room. Eleven to fourteen can hold the pencil. You
still hold the closed-tool rule. If you fail, say so. That honesty is the sitting. Shame is not the
sitting. Wild guesses are data. Missed catches are data. Reverse-engineered ballparks are
theater; start again.

For the Grown-Up Reader

Young-band booklet for *Number in the Age of AI*. Little 6/6 sittings are live; this prose follows those six, aged up: units, Photomath, headlines, Fermi size. It is not *Mathematics for Homeschooling*. It is not a philosopher booklet.

Sources: Pólya, *How to Solve It* (Princeton, 1945), especially the first questions of the heuristic. Feynman, “Cargo Cult Science,” Caltech commencement, 1974. Fermi is named for order-of-magnitude practice; this band does not quote a primary Fermi paper. Academy lines are labeled as sittings.

Hard edges from the young sitting still hold. No live timed test as the demo if it tempts cheating. Units matter. CAS and Photomath are in scope. Accommodations are access, not a ghost. Cite the tool if it changed a step. Medical dosing: professional, not a model. Do not let a model be the first check on a news statistic.

The six slugs: `estimate-first`, `path-on-paper`, `check-the-wild-answer`, `same-quantity-two-ways`, `about-not-exact`, `catch-the-machine`. Liam maps when this pack is actually live. Do not invent sittings 2–6 for young if the site only has Little on those slugs — sitting 1 (`estimate-first`) already exists in young. The booklet still serves the unit.

Do not convert this file until it is over 45 KB and the sister bands are over 60. Disk size is not gold. Live curl is gold. I will not ship a thin PDF and call it a booklet. Clare already taught that lesson. Bonaventure’s live `.md` under floor is the other warning. This pack waits until all four bands meet the floor.

NCTM has treated estimation as a standard, not a warm-up game, for decades. This booklet agrees. Estimation is how a citizen keeps an oracle honest. It is not a cute extra. If a school’s pacing guide skips it because a calculator is faster, the sitting still runs at the table. The calculator can wait. So can the model. The path does not wait — it has to be made.

About WisdomForge

WisdomForge writes the same idea four times: elementary, middle, high, and adult. The facts do not change. The sentences do. This is the middle band of *Number in the Age of AI*. Sittings stay live. This booklet is the download. Payment is pulled. There is no purchase CTA. All booklets are free. When the high band exists, it will treat the same six beats as academic integrity. When the adult band exists, it will treat them as a household alarm. Neither is this file. Do not convert a middle draft at 44 KB and call it gold. Forty-five is the floor. This sentence exists so the file actually crosses it instead of hovering at forty-four like a costume exact. Need both the size and the path. The sitting still counts.